Experiment.4

Aim: Hands-on Solidity Programming Assignments for creating Smart Contracts

Theory:

Solidity and Smart Contract Fundamentals
1. Primitive Data Types, Variables, and Functions (pure, view)

Solidity is a high-level, statically typed programming language used for developing smart contracts on the Ethereum blockchain. Primitive data types serve as the fundamental building blocks of Solidity and play a crucial role in defining variables, performing computations, and managing data within a contract.

Commonly used primitive data types include:

- **uint / int**: Used to represent unsigned and signed integers of different sizes, such as uint256 or int128. These are widely used for financial calculations, counters, and token balances.
- **bool**: Represents logical values (true or false) and is commonly used in decision-making statements such as access control and validations.
- **address**: Stores a 20-byte Ethereum account or contract address, which is essential for identifying users and interacting with other contracts.
- **bytes / string**: Used for storing binary and textual data, respectively, such as user names or encrypted information.

Variables in Solidity are categorized into three types based on their scope and lifetime:

- **State Variables**: Stored permanently on the blockchain and consume gas when modified. Example: storing the contract owner's address.
- **Local Variables**: Temporary variables that exist only during function execution and are not stored on-chain.
- **Global Variables**: Predefined variables such as msg.sender, msg.value, and block.timestamp, which provide information about the transaction and blockchain environment.

Functions in Solidity define the logic and behavior of smart contracts. Two important function types include:

- **pure**: Functions that neither read nor modify blockchain state; they operate only on input values. Example: a function that adds two numbers.
- **view**: Functions that can read state variables but cannot modify them. Example: a function that returns an account balance.

These function types help in optimizing gas usage and ensuring secure and predictable contract execution.

2. Inputs and Outputs to Functions

Functions in Solidity can accept input parameters and return output values, enabling interaction between users and smart contracts. Inputs allow external users or other contracts to pass data into a function, while outputs enable the function to return computed results.

For example, a function may accept an amount of Ether as input and return a boolean value indicating whether a transaction was successful. Solidity also supports named return variables, which enhance code readability and simplify debugging.

3. Visibility, Modifiers, and Constructors

Function visibility defines who can access a particular function within or outside the contract:

- **public**: Accessible both internally and externally.
- **private**: Accessible only within the same contract.
- **internal**: Accessible within the contract and its derived contracts.
- **external**: Can only be called by external accounts or other contracts.

Modifiers are special functions that modify the behavior of other functions. They are commonly used for security and access control. For instance, an onlyOwner modifier can restrict certain functions so that only the contract owner can execute them.

A **constructor** is a special function that executes only once during contract deployment. It is primarily used to initialize important variables, such as assigning the deploying account as the contract owner.

4. Control Flow: if-else and Loops

Solidity supports standard control flow structures similar to traditional programming languages.

- **if-else statements** allow conditional decision-making in contract logic. For example, a contract can verify whether a user has sufficient balance before processing a transaction.

- **Loops (for, while, do-while)** enable repeated execution of code, such as iterating through an array of registered users. However, loops must be used carefully, as excessive iterations increase gas consumption and may make transactions costly or even fail due to gas limits.

5. Data Structures: Arrays, Mappings, Structs, and Enums

Solidity provides several data structures to efficiently manage and organize data within smart contracts.

- **Arrays**: Used to store ordered collections of elements. They can be fixed-size or dynamic. Example: an array storing addresses of registered users.
- **Mappings**: Key-value storage structures that allow fast data retrieval. Example: mapping(address => uint) to store user balances. Unlike arrays, mappings do not support iteration.

Structs: Allow grouping of related data into a single entity. Example:

```
 struct Player {
    string
    name;    uint
    score;
}
```

- This structure helps store multiple attributes of a player in an organized manner.

Enums: Used to define a set of predefined constants, improving code clarity and readability. Example:

```
 enum Status { Pending, Active, Closed }
```

6. Data Locations

Solidity defines three primary data locations that determine where variables are stored and how they behave in memory:

- storage: Permanent data stored on the blockchain. Used for state variables.
- memory: Temporary data that exists only during function execution. Used for local variables and function inputs.
- calldata: A non-modifiable and non-persistent location used for external function parameters. It is more gas-efficient than memory.

Understanding data locations is essential because they directly impact gas costs, efficiency, and contract performance.

7. Transactions: Ether, Wei, Gas, and Sending

Transactions Ether and Wei:

Ether is the native cryptocurrency of the Ethereum blockchain. The smallest unit of Ether is Wei, where:

1 Ether = 10^{18} Wei.

This ensures high precision in financial and smart contract transactions.

Gas and Gas Price:

Every transaction on Ethereum consumes gas, which represents the computational cost of executing operations. The gas price determines how much Ether is paid per unit of gas. A higher gas price results in faster transaction processing.

Sending Transactions:

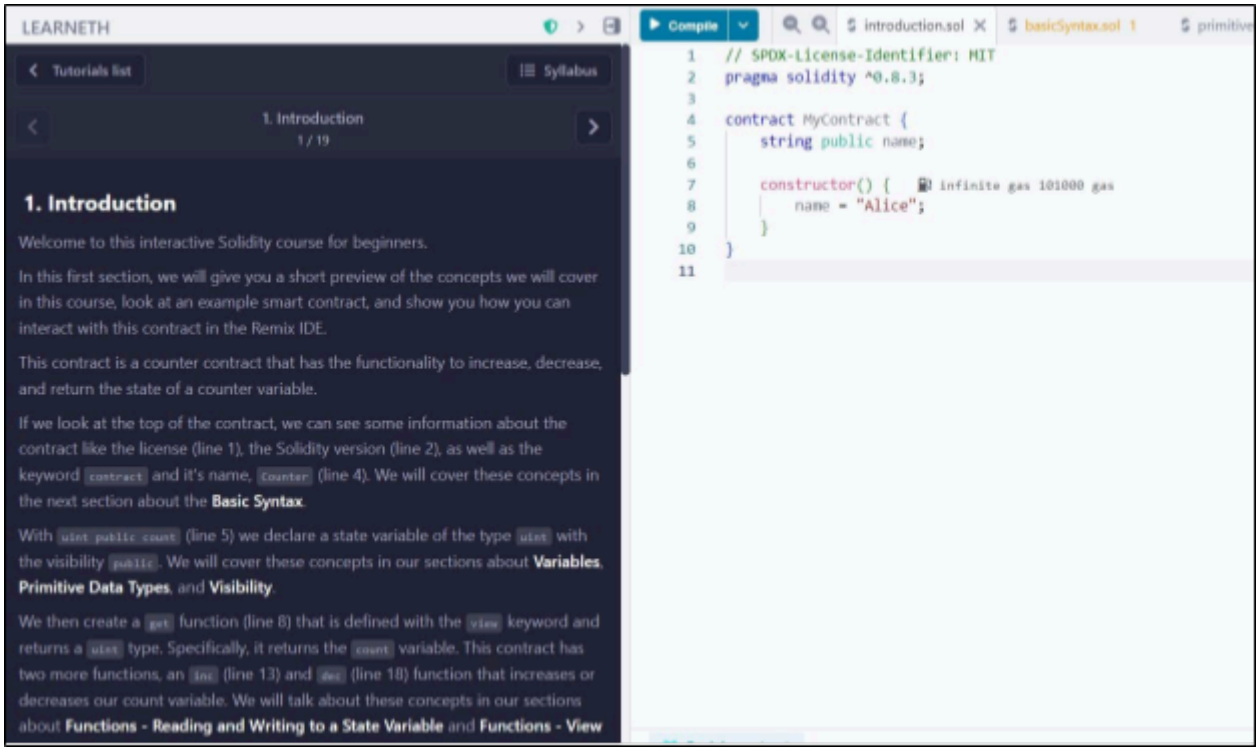
Transactions are used to transfer Ether or interact with smart contracts. Common methods include:

- transfer() – Safe but limited in gas usage.
- send() – Returns a boolean value indicating success or failure.
- call() – More flexible and recommended for modern Solidity development.

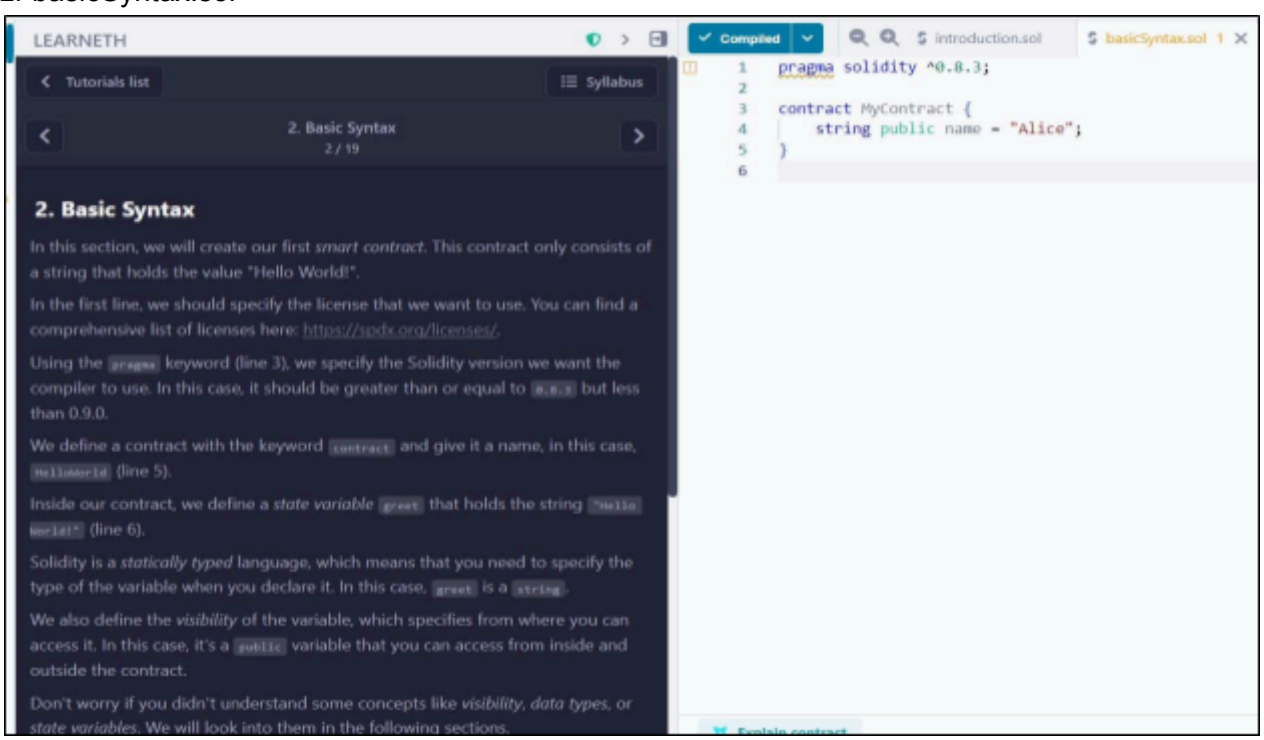
Efficient contract design is necessary to minimize gas consumption and reduce transaction costs.

Output -

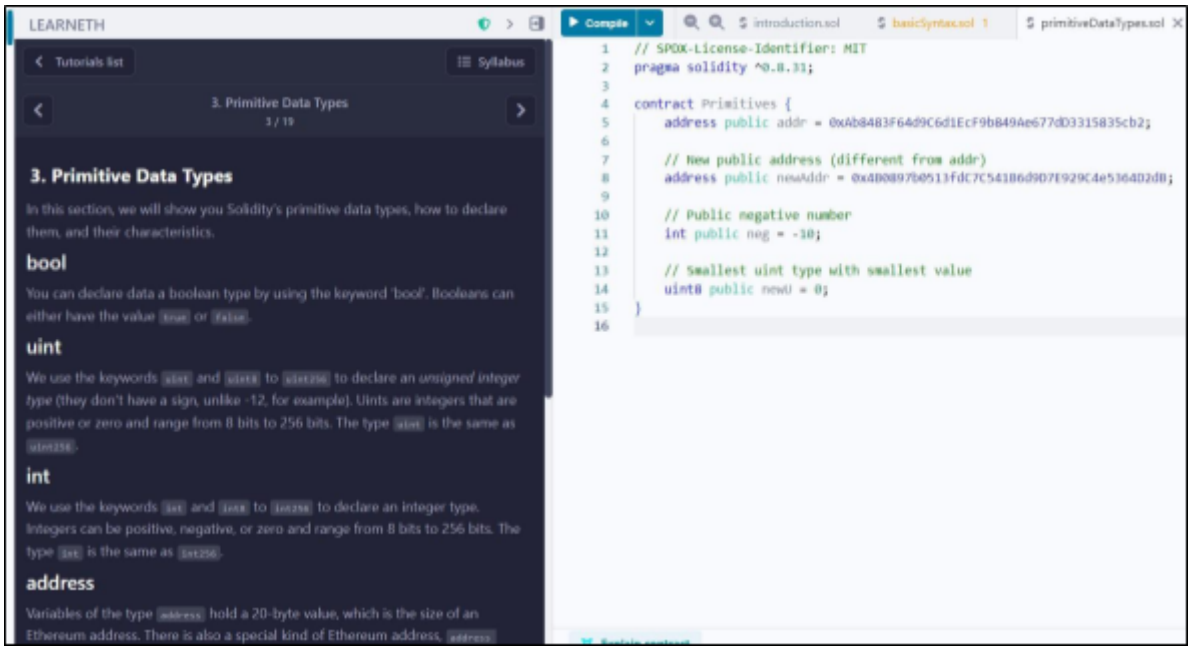
1.introduction.sol



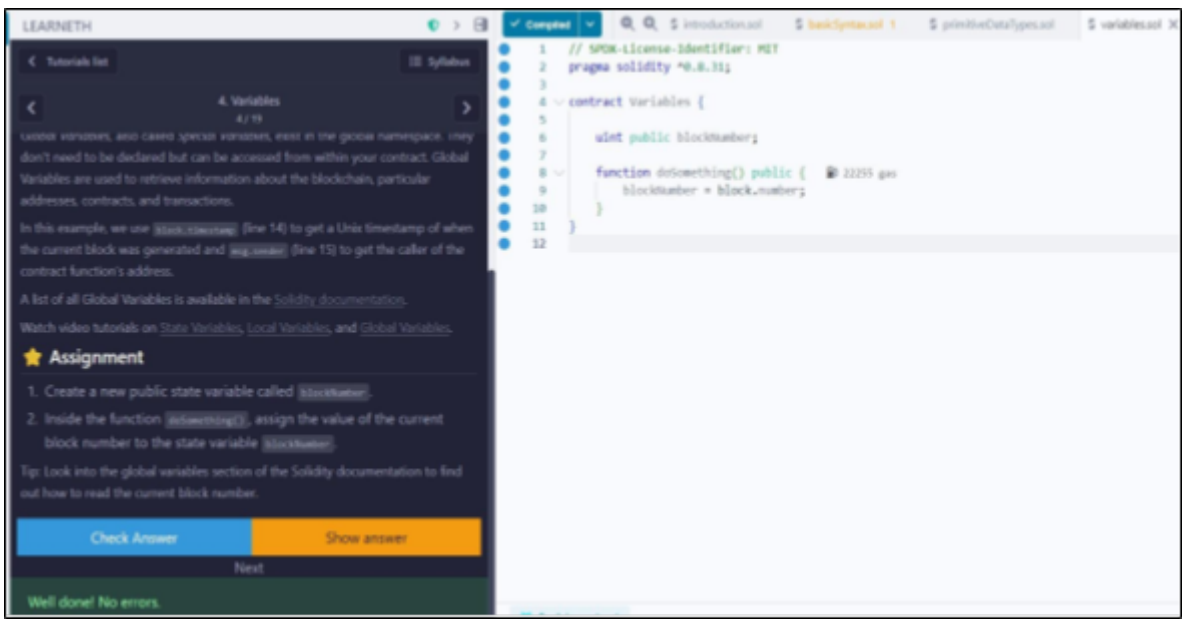
2. basicSyntax.sol



3. primitiveDataTypes.sol



4. variables.sol



5.1 readAndWrite.sol

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5.1 Functions - Reading and Writing to a State Variable
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5.1 Functions - Reading and Writing to a State Variable

This section will give a short introduction to functions and teach you how to use them to read from and write to a state variable.

As in other languages, we use functions in Solidity to create modular, reusable code. However, Solidity functions have some particularities.

Solidity functions can be split into two types:

1. Functions that modify the state of the blockchain, like writing to a state variable. In this contract, the `set` function (line 9) changes the state variable `num`.
2. Functions that don't modify the state of the blockchain. These functions are marked `view` or `pure`. For example, in this contract, the `get` function (line 14) marked `view` that only returns `num` does not change the state.

To define a function, use the `function` keyword followed by a unique name.

If the function takes inputs like our `set` function (line 9), you must specify the parameter types and names. A common convention is to use an underscore as a prefix for the parameter name to distinguish them from state variables.

You can then set the visibility of a function and declare them `view` or `pure` as we

Compile | variables.sol | .prettierrc.json | readAndWrite.sol

```
1  // SPDX-License-Identifier: MIT
2  pragma solidity ^0.8.31;
3
4  contract SimpleStorage {
5
6      bool public b = true;
7
8      function get_b() public view returns (bool) {    2472 gas
9          return b;
10     }
11 }
12
```

5.2 viewAndPure.sol

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5.2 Functions - View and Pure
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5.2 Functions - View and Pure

This section will look into the types of functions that don't modify the state of the blockchain: *view* and *pure* functions.

View Functions

View functions promise to not modify the state.

"The following statements are considered modifying the state:

1. Writing to state variables.
2. Emitting events.
3. Creating other contracts.
4. Using selfdestruct.
5. Sending Ether via calls.
6. Calling any function not marked view or pure.
7. Using low-level calls.
8. Using inline assembly that contains certain opcodes."

From the Solidity documentation.

You can declare a view function using the keyword `view`. In this contract, `addTo2` (line 8) is a view function. This function takes the parameter `y` and returns the

Compile | variables.sol | .prettierrc.json | readAndWrite.sol | viewAndPure.sol

```
1  // SPDX-License-Identifier: MIT
2  pragma solidity ^0.8.3;
3
4  contract ViewAndPure {
5      uint public x = 1;
6
7      // Promise not to modify the state.
8      function addTo2(uint y) public view returns (uint) {    infinite gas
9          return x + y;
10     }
11
12     // Promise not to modify or read from the state.
13     function add(uint i, uint j) public pure returns (uint) {    infinite gas
14         return i + j;
15     }
16
17     function addTo2(uint y) public {    infinite gas
18         x = x + y;
19     }
20 }
```

Deploy contract

5.3 modifiersAndConstructor.sol

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5.3 Functions - Modifiers and Constructors
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A constructor function is executed upon the creation of a contract. You can use it to run contract initialization code. The constructor can have parameters and is especially useful when you don't know certain initialization values before the deployment of the contract.

You declare a constructor using the `constructor` keyword. The constructor in this contract (line 11) sets the initial value of the owner variable upon the creation of the contract.

Watch a video tutorial on Function Modifiers.

Assignment

1. Create a new function, `increaseX` in the contract. The function should take an input parameter of type `uint` and increase the value of the variable `x` by the value of the input parameter.
2. Make sure that x can only be increased.
3. The body of the function `increaseX` should be empty.

Tip: Use modifiers.

Check Answer | Show answer

Next

Well done! No errors.

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```
1  // SPDX-License-Identifier: MIT
2  pragma solidity ^0.8.3;
3
4  contract FunctionModifier {
5      // We will use these variables to demonstrate how to use
6      // modifiers.
7      address public owner;
8      uint public x = 10;
9      bool public locked;
10
11     constructor() {    461217 gas 414400 gas
12         // Set the transaction sender as the owner of the contract.
13         owner = msg.sender;
14     }
15
16     // Modifier to check that the caller is the owner of
17     // the contract.
18     modifier onlyOwner() {
19         require(msg.sender == owner, "Not owner");
20         // Underscore is a special character only used inside
21         // a function modifier and it tells Solidity to
22         // execute the rest of the code.
23         _;
24     }
25
26     // Modifiers can take inputs. This modifier checks that the
27     // address passed in is not the zero address.
28     modifier validAddress(address _addr) {
29         require(_addr != address(0), "Not valid address");
30         _;
31     }
32
```

5.4 inputsAndOutputs.sol

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5.4 Functions - Inputs and Outputs
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5.4 Functions - Inputs and Outputs

In this section, we will learn more about the inputs and outputs of functions.

Multiple named Outputs

Functions can return multiple values that can be named and assigned to their name.

The `returnMany` function (line 6) shows how to return multiple values. You will often return multiple values. It could be a function that collects outputs of various functions and returns them in a single function call for example.

The `named` function (line 19) shows how to name return values. Naming return values helps with the readability of your contracts. Named return values make it easier to keep track of the values and the order in which they are returned. You can also assign values to a name.

The `assigned` function (line 33) shows how to assign values to a name. When you assign values to a name you can omit (leave out) the return statement and return them individually.

Deconstructing Assignments

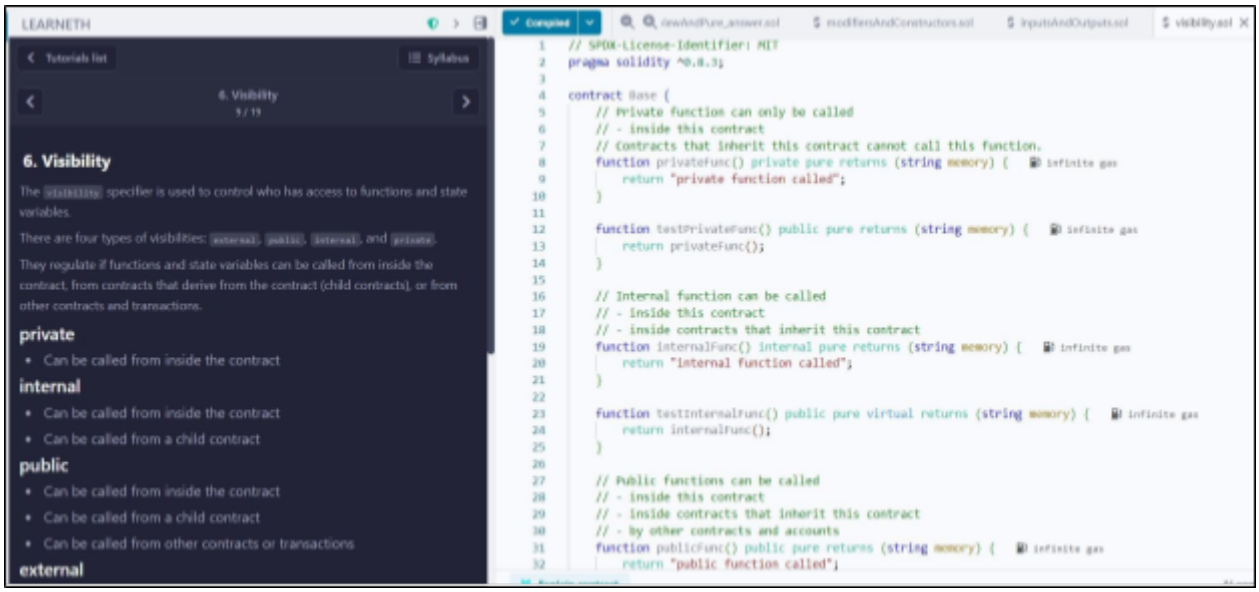
You can use deconstructing assignments to unpack values into distinct variables.

The `destructingAssignments` function (line 49) assigns the values of the `returnMany`

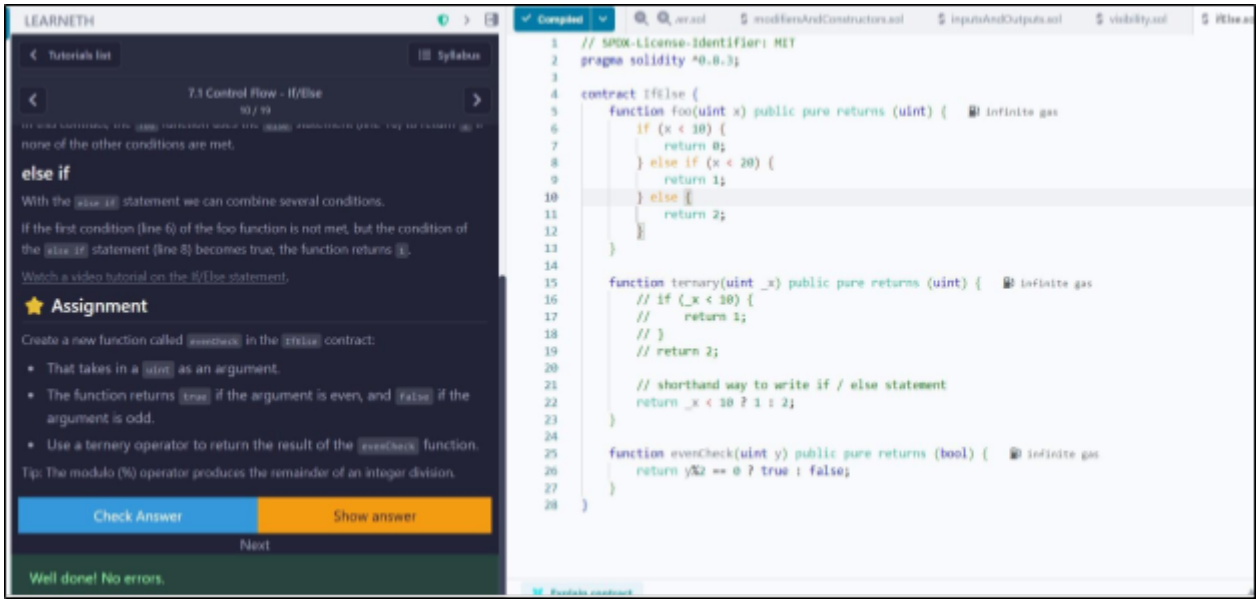
Compiled | ndPure.sol | viewAndPure_answer.sol | modifiersAndConstructors.sol | inputsAndOutputs.sol

```
1  // SPDX-License-Identifier: MIT
2  pragma solidity ^0.8.3;
3
4  contract Function {
5      // Functions can return multiple values.
6      function returnMany()    infinite gas
7          public
8          pure
9          returns (
10             uint,
11             bool,
12             uint
13         )
14     {
15         return (1, true, 2);
16     }
17
18     // Return values can be named.
19     function named()    infinite gas
20         public
21         pure
22         returns (
23             uint x,
24             bool b,
25             uint y
26         )
27     {
28         return (1, true, 2);
29     }
30
31     // Return values can be assigned to their name.
32     // In this case the return statement can be omitted.
```

6. visibility.sol



7.1 ControlFlow.sol



7.2 loops.sol

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7.2 Control Flow - Loops
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7.2 Control Flow - Loops

Solidity supports iterative control flow statements that allow contracts to execute code repeatedly.

Solidity differentiates between three types of loops: `for`, `while`, and `do while` loops.

for

Generally, `for` loops (line 7) are great if you know how many times you want to execute a certain block of code. In solidity, you should specify this amount to avoid transactions running out of gas and failing if the amount of iterations is too high.

while

If you don't know how many times you want to execute the code but want to break the loop based on a condition, you can use a `while` loop (line 20). Loops are seldom used in Solidity since transactions might run out of gas and fail if there is no limit to the number of iterations that can occur.

do while

The `do while` loop is a special kind of while loop where you can ensure the code is executed at least once, before checking on the condition.

```
1  // SPDX-License-Identifier: MIT
2  pragma solidity ^0.8.3;
3
4  contract Loop {
5      uint public count;
6      function loop() public{    infinite gas
7          // for loop
8          for (uint i = 0; i < 10; i++) {
9              if (i == 5) {
10                 // Skip to next iteration with continue
11                 continue;
12             }
13             if (i == 5) {
14                 // Exit loop with break
15                 break;
16             }
17             count++;
18         }
19
20         // while loop
21         uint j;
22         while (j < 10) {
23             j++;
24         }
25     }
26 }
27
```

8.1 arrays.sol

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8.1 Data Structures - Arrays
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8.1 Data Structures - Arrays

In the next sections, we will look into the data structures that we can use to organize and store our data in Solidity.

Arrays, mappings and structs are all reference types. Unlike value types (e.g. booleans or integers) reference types don't store their value directly. Instead, they store the location where the value is being stored. Multiple reference type variables could reference the same location, and a change in one variable would affect the others, therefore they need to be handled carefully.

In Solidity, an array stores an ordered list of values of the same type that are indexed numerically.

There are two types of arrays, compile-time *fixed-size* and *dynamic arrays*. For fixed-size arrays, we need to declare the size of the array before it is compiled. The size of dynamic arrays can be changed after the contract has been compiled.

Declaring arrays

We declare a fixed-size array by providing its type, array size (as an integer in square brackets), visibility, and name (line 9).

We declare a dynamic array in the same manner. However, we don't provide an array size and leave the brackets empty (line 6).

```
1  // SPDX-License-Identifier: MIT
2  pragma solidity ^0.8.3;
3
4  contract Array {
5      // Several ways to initialize an array
6      uint[] public arr;
7      uint[] public arr2 = [1, 2, 3];
8      // Fixed sized array, all elements initialize to 0
9      uint[10] public myFixedSizeArr;
10     uint[3] public arr3 = [0, 1, 2];
11
12     function get(uint i) public view returns (uint) {    infinite gas
13         return arr[i];
14     }
15
16     // Solidity can return the entire array.
17     // But this function should be avoided for
18     // arrays that can grow indefinitely in length.
19     function getArr() public view returns (uint[3] memory) {    infinite gas
20         return arr3;
21     }
22
23     function push(uint i) public {    46820 gas
24         // Append to array
25         // This will increase the array length by 1.
26         arr.push(i);
27     }
28
29     function pop() public {    20462 gas
30         // Remove last element from array
31         // This will decrease the array length by 1
32         arr.pop();
```

8.2 mapping.sol

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8.2 Data Structures - Mappings
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8.2 Data Structures - Mappings

In Solidity, mappings are a collection of key types and corresponding value type pairs.

The biggest difference between a mapping and an array is that you can't iterate over mappings. If we don't know a key we won't be able to access its value. If we need to know all of our data or iterate over it, we should use an array.

If we want to retrieve a value based on a known key we can use a mapping (e.g. addresses are often used as keys). Looking up values with a mapping is easier and cheaper than iterating over arrays. If arrays become too large, the gas cost of iterating over it could become too high and cause the transaction to fail.

We could also store the keys of a mapping in an array that we can iterate over.

Creating mappings

Mappings are declared with the syntax `mapping(KeyType => ValueType) VariableName`. The key type can be any built-in value type or any contract, but not a reference type. The value type can be of any type.

In this contract, we are creating the public mapping `myMap` (line 6) that associates the key type `address` with the value type `uint`.

Accessing values

Compiled | vAndOutputs.sol | visibility.sol | iRline.sol | loops.sol | arrays.sol | mappings.sol

```
1  // SPDX-License-Identifier: MIT
2  pragma solidity ^0.8.3;
3
4  contract Mapping {
5      // Mapping from address to uint
6      mapping(address => uint) public balances;
7
8      function get(address _addr) public view returns (uint) {    2872 gas
9          // Mapping always returns a value.
10         // If the value was never set, it will return the default value.
11         return balances[_addr];
12     }
13
14     function set(address _addr) public {    25256 gas
15         // Update the value at this address
16         balances[_addr] = _addr.balance;
17     }
18
19     function remove(address _addr) public {    5566 gas
20         // Reset the value to the default value.
21         delete balances[_addr];
22     }
23 }
24
25 contract NestedMapping {
26     // Nested mapping (mapping from address to another mapping)
27     mapping(address => mapping(uint => bool)) public nested;
28
29     function get(address _addr1, uint _i) public view returns (bool) {    3159 gas
30         // You can get values from a nested mapping
31         // even when it is not initialized
32         return nested[_addr1][_i];
```

8.3 structs.sol

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8.3 Data Structures - Structs
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8.3 Data Structures - Structs

In Solidity, we can define custom data types in the form of structs. Structs are a collection of variables that can consist of different data types.

Defining structs

We define a struct using the `struct` keyword and a name (line 5). Inside curly braces, we can define our struct's members, which consist of the variable names and their data types.

Initializing structs

There are different ways to initialize a struct.

Positional parameters: We can provide the name of the struct and the values of its members as parameters in parentheses (line 16).

Key-value mapping: We provide the name of the struct and the keys and values as a mapping inside curly braces (line 19).

Initialize and update a struct: We initialize an empty struct first and then update its member by assigning it a new value (line 23).

Accessing structs

To access a member of a struct we can use the dot operator (line 33).

Compiled | visibility.sol | iRline.sol | loops.sol | arrays.sol | mappings.sol | structs.sol

```
1  // SPDX-License-Identifier: MIT
2  pragma solidity ^0.8.3;
3
4  contract Todos {
5      struct Todo {
6          string text;
7          bool completed;
8      }
9
10     // An array of 'todo' structs
11     Todo[] public todos;
12
13     function create(string memory _text) public {    infinite gas
14         // 3 ways to initialize a struct
15         // - calling it like a function
16         todos.push(Todo(_text, false));
17
18         // key value mapping
19         todos.push(Todo({text: _text, completed: false}));
20
21         // initialize an empty struct and then update it
22         Todo memory todo;
23         todo.text = _text;
24         // todo.completed initialized to false
25
26         todos.push(todo);
27     }
28
29     // Solidity automatically created a getter for 'todos' so
30     // you don't actually need this function.
31     function get(uint _index) public view returns (string memory text, bool completed) {    infinite gas
32         Todo storage todo = todos[_index];
```

8.4 enums.sol

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8.4 Data Structures - Enums
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8.4 Data Structures - Enums

In Solidity *enums* are custom data types consisting of a limited set of constant values. We use enums when our variables should only get assigned a value from a predefined set of values.

In this contract, the state variable `status` can get assigned a value from the limited set of provided values of the enum `status` representing the various states of a shipping status.

Defining enums

We define an enum with the enum keyword, followed by the name of the custom type we want to create (line 6). Inside the curly braces, we define all available members of the enum.

Initializing an enum variable

We can initialize a new variable of an enum type by providing the name of the enum, the visibility, and the name of the variable (line 16). Upon its initialization, the variable will be assigned the value of the first member of the enum, in this case, Pending (line 7).

Even though enum members are named when you define them, they are stored as unsigned integers, not strings. They are numbered in the order that they were defined, the first member starting at 0. The initial value of status, in this case, is 0.

Compiled | #1bin.sol | loops.sol | arrays.sol | mappings.sol | structs.sol | enums.sol

```
1  // SPDX-License-Identifier: MIT
2  pragma solidity ^0.8.3;
3
4  contract Enum {
5      // Enum representing shipping status
6      enum Status {
7          Pending,
8          Shipped,
9          Accepted,
10         Rejected,
11         Canceled
12     }
13
14     enum Size {
15         S,
16         M,
17         L
18     }
19
20     // Default value is the first element listed in
21     // definition of the type, in this case "Pending"
22     Status public status;
23     Size public sizes;
24
25     function get() public view returns (Status) {    2605 gas
26         return status;
27     }
28
29     function getSize() public view returns (Size) {    2633 gas
30         return sizes;
31     }
32
```

9. dataLocations.sol

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9. Data Locations
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9. Data Locations

The values of variables in Solidity can be stored in different data locations: *memory*, *storage*, and *calldata*.

As we have discussed before, variables of the value type store an independent copy of a value, while variables of the reference type (array, struct, mapping) only store the location (reference) of the value.

If we use a reference type in a function, we have to specify in which data location their values are stored. The price for the execution of the function is influenced by the data location; creating copies from reference types costs gas.

Storage

Values stored in *storage* are stored permanently on the blockchain and, therefore, are expensive to use.

In this contract, the state variables `arr`, `map`, and `myStructs` (lines 5, 6, and 10) are stored in storage. State variables are always stored in storage.

Memory

Values stored in *memory* are only stored temporarily and are not on the blockchain. They only exist during the execution of an external function and are discarded afterward. They are cheaper to use than values stored in *storage*.

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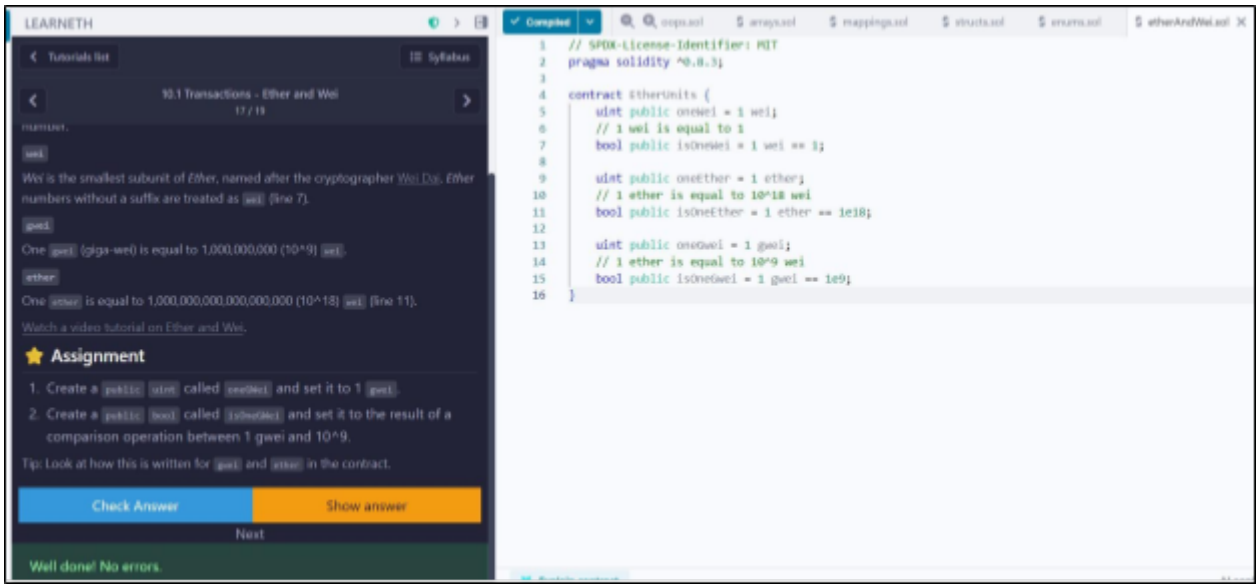
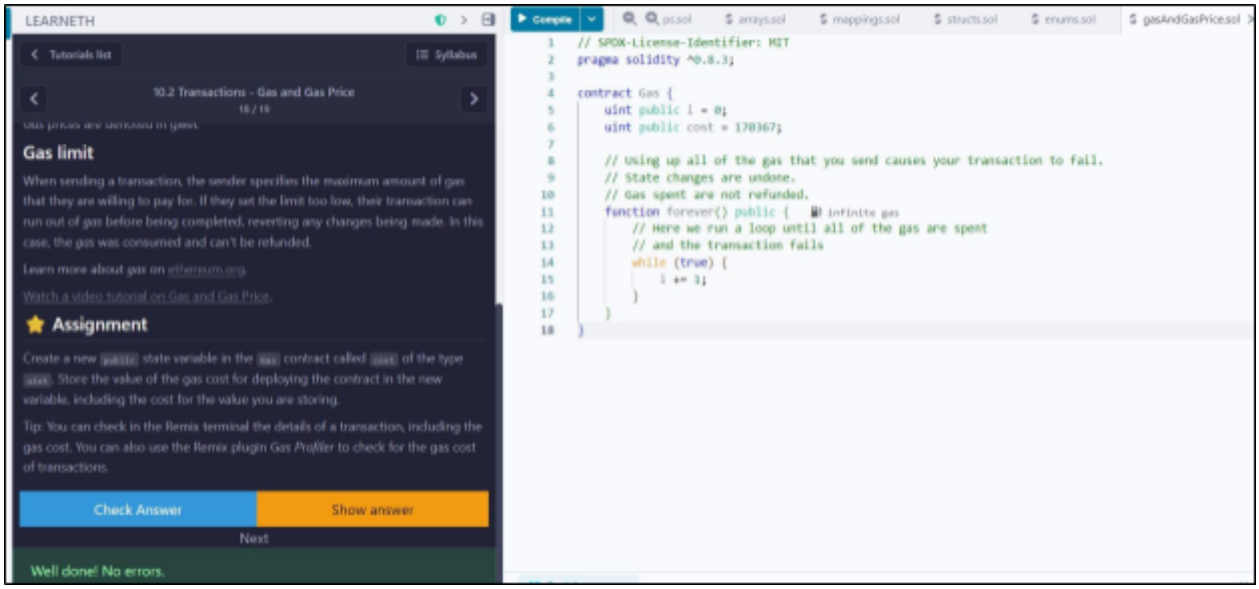
```
1  // SPDX-License-Identifier: MIT
2  pragma solidity ^0.8.3;
3
4  contract DataLocations {
5      uint[] public arr;
6      mapping(uint => address) map;
7      struct MyStruct {
8          uint foo;
9      }
10     mapping(uint => MyStruct) public myStructs;
11
12     function f() public returns (MyStruct memory, MyStruct memory, MyStruct memory){    infinite gas
13         // call _f with state variables
14         _f(arr, map, myStructs[1]);
15         // get a struct from a mapping
16         MyStruct storage myStruct = myStructs[1];
17         myStruct.foo = 4;
18         // create a struct in memory
19         MyStruct memory myMemStruct = MyStruct(0);
20         MyStruct memory myMemStruct2 = myMemStruct;
21         myMemStruct2.foo = 1;
22
23         MyStruct memory myMemStruct3 = myStruct;
24         myMemStruct3.foo = 3;
25         return (myStruct, myMemStruct2, myMemStruct3);
26     }
27
28     function _f(    undefined gas
29         uint[] storage _arr,
30         mapping(uint => address) storage _map,
31         MyStruct storage _myStruct
32     ) internal {
```

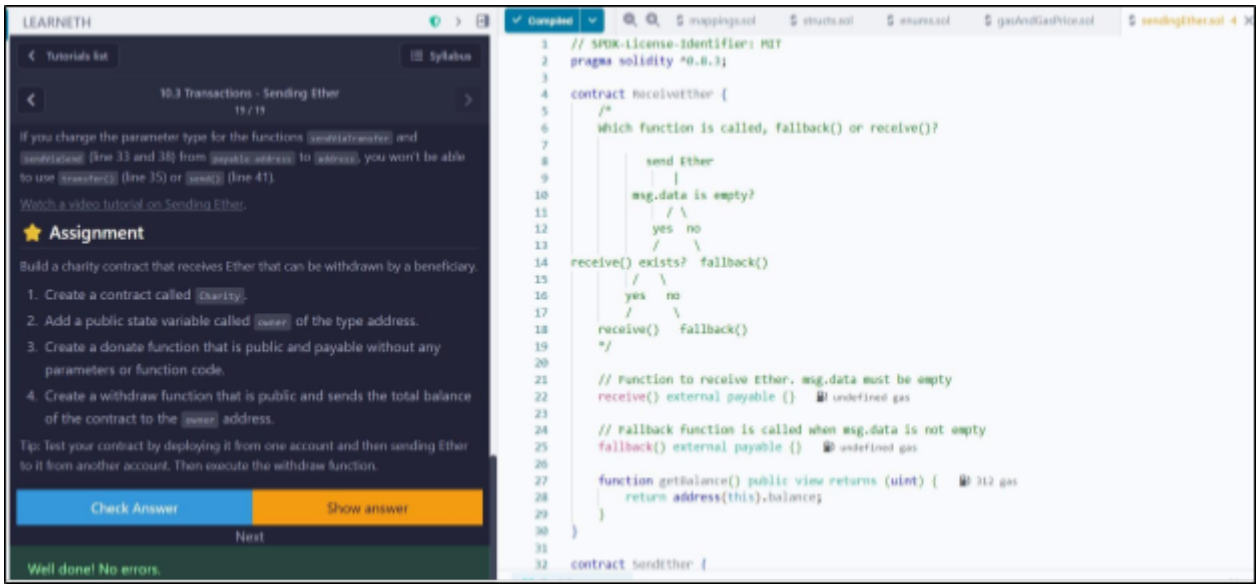
10.1 etherAndWei.sol



10.2 gasAndGasPrice.sol



10.3 sendingEther.sol



Conclusion - This experiment provides hands-on experience in Solidity programming by exploring core blockchain concepts, data types, control structures, and transaction handling. Understanding these fundamentals enables the design and deployment of secure, efficient, and reliable smart contracts on the Ethereum blockchain.